Interacting Multipoles of Rank 1 and 4:

$$T_{\alpha\beta\gamma\delta\epsilon}\mu_\alpha\Phi_{\beta\gamma\delta\epsilon} = R_z^{-11}\cdot\frac{1}{105}\cdot\left[\begin{array}{c}
-945\cdot R_\alpha\cdot R_\beta\cdot R_\gamma\cdot R_\delta\cdot R_\epsilon\cdot\mu_\alpha\cdot\Phi_{\beta\gamma\delta\epsilon}\\
+105\cdot R_z^2\cdot R_\alpha\cdot R_\beta\cdot R_\gamma\cdot\delta\left(\delta,\epsilon\right)\cdot\mu_\alpha\cdot\Phi_{\beta\gamma\delta\epsilon}\\
+105\cdot R_z^2\cdot R_\alpha\cdot R_\beta\cdot R_\delta\cdot\delta\left(\gamma,\epsilon\right)\cdot\mu_\alpha\cdot\Phi_{\beta\gamma\delta\epsilon}\\
+105\cdot R_z^2\cdot R_\alpha\cdot R_\beta\cdot R_\epsilon\cdot\delta\left(\gamma,\delta\right)\cdot\mu_\alpha\cdot\Phi_{\beta\gamma\delta\epsilon}\\
+105\cdot R_z^2\cdot R_\alpha\cdot R_\gamma\cdot R_\delta\cdot\delta\left(\beta,\epsilon\right)\cdot\mu_\alpha\cdot\Phi_{\beta\gamma\delta\epsilon}\\
+105\cdot R_z^2\cdot R_\alpha\cdot R_\gamma\cdot R_\epsilon\cdot\delta\left(\beta,\delta\right)\cdot\mu_\alpha\cdot\Phi_{\beta\gamma\delta\epsilon}\\
+105\cdot R_z^2\cdot R_\alpha\cdot R_\delta\cdot R_\epsilon\cdot\delta\left(\beta,\gamma\right)\cdot\mu_\alpha\cdot\Phi_{\beta\gamma\delta\epsilon}\\
+105\cdot R_z^2\cdot R_\beta\cdot R_\gamma\cdot R_\delta\cdot\delta\left(\alpha,\epsilon\right)\cdot\mu_\alpha\cdot\Phi_{\beta\gamma\delta\epsilon}\\
+105\cdot R_z^2\cdot R_\beta\cdot R_\gamma\cdot R_\epsilon\cdot\delta\left(\alpha,\delta\right)\cdot\mu_\alpha\cdot\Phi_{\beta\gamma\delta\epsilon}\\
+105\cdot R_z^2\cdot R_\beta\cdot R_\delta\cdot R_\epsilon\cdot\delta\left(\alpha,\gamma\right)\cdot\mu_\alpha\cdot\Phi_{\beta\gamma\delta\epsilon}\\
+105\cdot R_z^2\cdot R_\gamma\cdot R_\delta\cdot R_\epsilon\cdot\delta\left(\alpha,\beta\right)\cdot\mu_\alpha\cdot\Phi_{\beta\gamma\delta\epsilon}\\
-15\cdot R_z^4\cdot R_\alpha\cdot\delta\left(\beta,\gamma\right)\cdot\delta\left(\delta,\epsilon\right)\cdot\mu_\alpha\cdot\Phi_{\beta\gamma\delta\epsilon}\\
-15\cdot R_z^4\cdot R_\alpha\cdot\delta\left(\beta,\delta\right)\cdot\delta\left(\gamma,\epsilon\right)\cdot\mu_\alpha\cdot\Phi_{\beta\gamma\delta\epsilon}\\
-15\cdot R_z^4\cdot R_\alpha\cdot\delta\left(\beta,\epsilon\right)\cdot\delta\left(\gamma,\delta\right)\cdot\mu_\alpha\cdot\Phi_{\beta\gamma\delta\epsilon}\\
-15\cdot R_z^4\cdot R_\beta\cdot\delta\left(\alpha,\gamma\right)\cdot\delta\left(\delta,\epsilon\right)\cdot\mu_\alpha\cdot\Phi_{\beta\gamma\delta\epsilon}\\
-15\cdot R_z^4\cdot R_\beta\cdot\delta\left(\alpha,\delta\right)\cdot\delta\left(\gamma,\epsilon\right)\cdot\mu_\alpha\cdot\Phi_{\beta\gamma\delta\epsilon}\\
-15\cdot R_z^4\cdot R_\beta\cdot\delta\left(\alpha,\epsilon\right)\cdot\delta\left(\gamma,\delta\right)\cdot\mu_\alpha\cdot\Phi_{\beta\gamma\delta\epsilon}\\
-15\cdot R_z^4\cdot R_\gamma\cdot\delta\left(\alpha,\beta\right)\cdot\delta\left(\delta,\epsilon\right)\cdot\mu_\alpha\cdot\Phi_{\beta\gamma\delta\epsilon}\\
-15\cdot R_z^4\cdot R_\gamma\cdot\delta\left(\alpha,\delta\right)\cdot\delta\left(\beta,\epsilon\right)\cdot\mu_\alpha\cdot\Phi_{\beta\gamma\delta\epsilon}\\
-15\cdot R_z^4\cdot R_\gamma\cdot\delta\left(\alpha,\epsilon\right)\cdot\delta\left(\beta,\delta\right)\cdot\mu_\alpha\cdot\Phi_{\beta\gamma\delta\epsilon}\\
-15\cdot R_z^4\cdot R_\delta\cdot\delta\left(\alpha,\beta\right)\cdot\delta\left(\gamma,\epsilon\right)\cdot\mu_\alpha\cdot\Phi_{\beta\gamma\delta\epsilon}\\
-15\cdot R_z^4\cdot R_\delta\cdot\delta\left(\alpha,\gamma\right)\cdot\delta\left(\beta,\epsilon\right)\cdot\mu_\alpha\cdot\Phi_{\beta\gamma\delta\epsilon}\\
-15\cdot R_z^4\cdot R_\delta\cdot\delta\left(\alpha,\epsilon\right)\cdot\delta\left(\beta,\gamma\right)\cdot\mu_\alpha\cdot\Phi_{\beta\gamma\delta\epsilon}\\
-15\cdot R_z^4\cdot R_\epsilon\cdot\delta\left(\alpha,\beta\right)\cdot\delta\left(\gamma,\delta\right)\cdot\mu_\alpha\cdot\Phi_{\beta\gamma\delta\epsilon}\\
-15\cdot R_z^4\cdot R_\epsilon\cdot\delta\left(\alpha,\gamma\right)\cdot\delta\left(\beta,\delta\right)\cdot\mu_\alpha\cdot\Phi_{\beta\gamma\delta\epsilon}\\
-15\cdot R_z^4\cdot R_\epsilon\cdot\delta\left(\alpha,\delta\right)\cdot\delta\left(\beta,\gamma\right)\cdot\mu_\alpha\cdot\Phi_{\beta\gamma\delta\epsilon}
\end{array}\right] \tag{1}$$

Simplify deltas

$$T_{\alpha\beta\gamma\delta\epsilon}\mu_\alpha\Phi_{\beta\gamma\delta\epsilon} = R_z^{-11} \cdot \frac{1}{105} \cdot \left[\begin{array}{c} -945 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \mu_\alpha \cdot \Phi_{\beta\gamma\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot \delta\left(\delta,\delta\right) \cdot \mu_\alpha \cdot \Phi_{\beta\gamma\delta\delta} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot \delta\left(\gamma,\gamma\right) \cdot \mu_\alpha \cdot \Phi_{\beta\gamma\gamma\delta} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\epsilon \cdot \delta\left(\gamma,\gamma\right) \cdot \mu_\alpha \cdot \Phi_{\beta\gamma\gamma\epsilon} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot \delta\left(\beta,\beta\right) \cdot \mu_\alpha \cdot \Phi_{\beta\beta\gamma\delta} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\epsilon \cdot \delta\left(\beta,\beta\right) \cdot \mu_\alpha \cdot \Phi_{\beta\beta\gamma\epsilon} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot R_\epsilon \cdot \delta\left(\beta,\beta\right) \cdot \mu_\alpha \cdot \Phi_{\beta\beta\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \delta\left(\alpha,\alpha\right) \cdot \mu_\alpha \cdot \Phi_{\alpha\beta\gamma\delta} \\ +105 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\epsilon \cdot \delta\left(\alpha,\alpha\right) \cdot \mu_\alpha \cdot \Phi_{\alpha\beta\gamma\epsilon} \\ +105 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot R_\epsilon \cdot \delta\left(\alpha,\alpha\right) \cdot \mu_\alpha \cdot \Phi_{\alpha\beta\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \delta\left(\alpha,\alpha\right) \cdot \mu_\alpha \cdot \Phi_{\alpha\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_\alpha \cdot \delta\left(\beta,\beta\right) \cdot \delta\left(\delta,\delta\right) \cdot \mu_\alpha \cdot \Phi_{\beta\beta\delta\delta} \\ -15 \cdot R_z^4 \cdot R_\alpha \cdot \delta\left(\beta,\beta\right) \cdot \delta\left(\gamma,\gamma\right) \cdot \mu_\alpha \cdot \Phi_{\beta\beta\gamma\gamma} \\ -15 \cdot R_z^4 \cdot R_\alpha \cdot \delta\left(\beta,\beta\right) \cdot \delta\left(\gamma,\gamma\right) \cdot \mu_\alpha \cdot \Phi_{\beta\beta\gamma\gamma} \\ -15 \cdot R_z^4 \cdot R_\beta \cdot \delta\left(\alpha,\alpha\right) \cdot \delta\left(\delta,\delta\right) \cdot \mu_\alpha \cdot \Phi_{\alpha\beta\delta\delta} \\ -15 \cdot R_z^4 \cdot R_\beta \cdot \delta\left(\alpha,\alpha\right) \cdot \delta\left(\gamma,\gamma\right) \cdot \mu_\alpha \cdot \Phi_{\alpha\beta\gamma\gamma} \\ -15 \cdot R_z^4 \cdot R_\beta \cdot \delta\left(\alpha,\alpha\right) \cdot \delta\left(\gamma,\gamma\right) \cdot \mu_\alpha \cdot \Phi_{\alpha\beta\gamma\gamma} \\ -15 \cdot R_z^4 \cdot R_\gamma \cdot \delta\left(\alpha,\alpha\right) \cdot \delta\left(\delta,\delta\right) \cdot \mu_\alpha \cdot \Phi_{\alpha\gamma\delta\delta} \\ -15 \cdot R_z^4 \cdot R_\gamma \cdot \delta\left(\alpha,\alpha\right) \cdot \delta\left(\beta,\beta\right) \cdot \mu_\alpha \cdot \Phi_{\alpha\beta\beta\gamma} \\ -15 \cdot R_z^4 \cdot R_\gamma \cdot \delta\left(\alpha,\alpha\right) \cdot \delta\left(\beta,\beta\right) \cdot \mu_\alpha \cdot \Phi_{\alpha\beta\beta\gamma} \\ -15 \cdot R_z^4 \cdot R_\delta \cdot \delta\left(\alpha,\alpha\right) \cdot \delta\left(\gamma,\gamma\right) \cdot \mu_\alpha \cdot \Phi_{\alpha\gamma\gamma\delta} \\ -15 \cdot R_z^4 \cdot R_\delta \cdot \delta\left(\alpha,\alpha\right) \cdot \delta\left(\beta,\beta\right) \cdot \mu_\alpha \cdot \Phi_{\alpha\beta\beta\delta} \\ -15 \cdot R_z^4 \cdot R_\delta \cdot \delta\left(\alpha,\alpha\right) \cdot \delta\left(\beta,\beta\right) \cdot \mu_\alpha \cdot \Phi_{\alpha\beta\beta\delta} \\ -15 \cdot R_z^4 \cdot R_\epsilon \cdot \delta\left(\alpha,\alpha\right) \cdot \delta\left(\gamma,\gamma\right) \cdot \mu_\alpha \cdot \Phi_{\alpha\gamma\gamma\epsilon} \\ -15 \cdot R_z^4 \cdot R_\epsilon \cdot \delta\left(\alpha,\alpha\right) \cdot \delta\left(\beta,\beta\right) \cdot \mu_\alpha \cdot \Phi_{\alpha\beta\beta\epsilon} \\ -15 \cdot R_z^4 \cdot R_\epsilon \cdot \delta\left(\alpha,\alpha\right) \cdot \delta\left(\beta,\beta\right) \cdot \mu_\alpha \cdot \Phi_{\alpha\beta\beta\epsilon} \end{array} \right] \tag{2a}$$

$$\xRightarrow{\text{cleaned up:}} R_z^{-11} \cdot \frac{1}{105} \cdot \left[\begin{array}{c} -945 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \mu_\alpha \cdot \Phi_{\beta\gamma\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot \mu_\alpha \cdot \Phi_{\beta\gamma\delta\delta} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot \mu_\alpha \cdot \Phi_{\beta\gamma\gamma\delta} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\epsilon \cdot \mu_\alpha \cdot \Phi_{\beta\gamma\gamma\epsilon} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot \mu_\alpha \cdot \Phi_{\beta\beta\gamma\delta} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\epsilon \cdot \mu_\alpha \cdot \Phi_{\beta\beta\gamma\epsilon} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot R_\epsilon \cdot \mu_\alpha \cdot \Phi_{\beta\beta\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \mu_\alpha \cdot \Phi_{\alpha\beta\gamma\delta} \\ +105 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\epsilon \cdot \mu_\alpha \cdot \Phi_{\alpha\beta\gamma\epsilon} \\ +105 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot R_\epsilon \cdot \mu_\alpha \cdot \Phi_{\alpha\beta\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \mu_\alpha \cdot \Phi_{\alpha\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_\alpha \cdot \mu_\alpha \cdot \Phi_{\beta\beta\delta\delta} \\ -15 \cdot R_z^4 \cdot R_\alpha \cdot \mu_\alpha \cdot \Phi_{\beta\beta\gamma\gamma} \\ -15 \cdot R_z^4 \cdot R_\alpha \cdot \mu_\alpha \cdot \Phi_{\beta\beta\gamma\gamma} \\ -15 \cdot R_z^4 \cdot R_\beta \cdot \mu_\alpha \cdot \Phi_{\alpha\beta\delta\delta} \\ -15 \cdot R_z^4 \cdot R_\beta \cdot \mu_\alpha \cdot \Phi_{\alpha\beta\gamma\gamma} \\ -15 \cdot R_z^4 \cdot R_\beta \cdot \mu_\alpha \cdot \Phi_{\alpha\beta\gamma\gamma} \\ -15 \cdot R_z^4 \cdot R_\gamma \cdot \mu_\alpha \cdot \Phi_{\alpha\gamma\delta\delta} \\ -15 \cdot R_z^4 \cdot R_\gamma \cdot \mu_\alpha \cdot \Phi_{\alpha\beta\beta\gamma} \\ -15 \cdot R_z^4 \cdot R_\gamma \cdot \mu_\alpha \cdot \Phi_{\alpha\beta\beta\gamma} \\ -15 \cdot R_z^4 \cdot R_\delta \cdot \mu_\alpha \cdot \Phi_{\alpha\gamma\gamma\delta} \\ -15 \cdot R_z^4 \cdot R_\delta \cdot \mu_\alpha \cdot \Phi_{\alpha\beta\beta\delta} \\ -15 \cdot R_z^4 \cdot R_\delta \cdot \mu_\alpha \cdot \Phi_{\alpha\beta\beta\delta} \\ -15 \cdot R_z^4 \cdot R_\epsilon \cdot \mu_\alpha \cdot \Phi_{\alpha\gamma\gamma\epsilon} \\ -15 \cdot R_z^4 \cdot R_\epsilon \cdot \mu_\alpha \cdot \Phi_{\alpha\beta\beta\epsilon} \\ -15 \cdot R_z^4 \cdot R_\epsilon \cdot \mu_\alpha \cdot \Phi_{\alpha\beta\beta\epsilon} \end{array} \right] \tag{2b}$$

Simplify R

$$T_{\alpha\beta\gamma\delta\epsilon}\mu_\alpha\Phi_{\beta\gamma\delta\epsilon} = R_z^{-11}\cdot\frac{1}{105}\cdot\begin{bmatrix} -945\cdot R_z\cdot R_z\cdot R_z\cdot R_z\cdot R_z\cdot\mu_z\cdot\Phi_{zzzz} \\ +105\cdot R_z^2\cdot R_z\cdot R_z\cdot R_z\cdot\mu_z\cdot\Phi_{zz\delta\delta} \\ +105\cdot R_z^2\cdot R_z\cdot R_z\cdot R_z\cdot\mu_z\cdot\Phi_{zz\gamma\gamma} \\ +105\cdot R_z^2\cdot R_z\cdot R_z\cdot R_z\cdot\mu_z\cdot\Phi_{zz\gamma\gamma} \\ +105\cdot R_z^2\cdot R_z\cdot R_z\cdot R_z\cdot\mu_z\cdot\Phi_{zz\beta\beta} \\ +105\cdot R_z^2\cdot R_z\cdot R_z\cdot R_z\cdot\mu_z\cdot\Phi_{zz\beta\beta} \\ +105\cdot R_z^2\cdot R_z\cdot R_z\cdot R_z\cdot\mu_z\cdot\Phi_{zz\beta\beta} \\ +105\cdot R_z^2\cdot R_z\cdot R_z\cdot R_z\cdot\mu_\alpha\cdot\Phi_{zzz\alpha} \\ +105\cdot R_z^2\cdot R_z\cdot R_z\cdot R_z\cdot\mu_\alpha\cdot\Phi_{zzz\alpha} \\ +105\cdot R_z^2\cdot R_z\cdot R_z\cdot R_z\cdot\mu_\alpha\cdot\Phi_{zzz\alpha} \\ +105\cdot R_z^2\cdot R_z\cdot R_z\cdot R_z\cdot\mu_\alpha\cdot\Phi_{zzz\alpha} \\ -15\cdot R_z^4\cdot R_z\cdot\mu_z\cdot\Phi_{\beta\beta\delta\delta} \\ -15\cdot R_z^4\cdot R_z\cdot\mu_z\cdot\Phi_{\beta\beta\gamma\gamma} \\ -15\cdot R_z^4\cdot R_z\cdot\mu_z\cdot\Phi_{\beta\beta\gamma\gamma} \\ -15\cdot R_z^4\cdot R_z\cdot\mu_\alpha\cdot\Phi_{z\alpha\delta\delta} \\ -15\cdot R_z^4\cdot R_z\cdot\mu_\alpha\cdot\Phi_{z\alpha\gamma\gamma} \\ -15\cdot R_z^4\cdot R_z\cdot\mu_\alpha\cdot\Phi_{z\alpha\gamma\gamma} \\ -15\cdot R_z^4\cdot R_z\cdot\mu_\alpha\cdot\Phi_{z\alpha\delta\delta} \\ -15\cdot R_z^4\cdot R_z\cdot\mu_\alpha\cdot\Phi_{z\alpha\beta\beta} \\ -15\cdot R_z^4\cdot R_z\cdot\mu_\alpha\cdot\Phi_{z\alpha\beta\beta} \\ -15\cdot R_z^4\cdot R_z\cdot\mu_\alpha\cdot\Phi_{z\alpha\gamma\gamma} \\ -15\cdot R_z^4\cdot R_z\cdot\mu_\alpha\cdot\Phi_{z\alpha\beta\beta} \\ -15\cdot R_z^4\cdot R_z\cdot\mu_\alpha\cdot\Phi_{z\alpha\beta\beta} \\ -15\cdot R_z^4\cdot R_z\cdot\mu_\alpha\cdot\Phi_{z\alpha\gamma\gamma} \\ -15\cdot R_z^4\cdot R_z\cdot\mu_\alpha\cdot\Phi_{z\alpha\beta\beta} \\ -15\cdot R_z^4\cdot R_z\cdot\mu_\alpha\cdot\Phi_{z\alpha\beta\beta} \end{bmatrix} \tag{3a}$$

$$\xRightarrow{\text{cleaned up:}} R_z^{-11}\cdot\frac{1}{105}\cdot\begin{bmatrix} +105\cdot R_z^5\cdot\mu_\alpha\cdot\Phi_{zzz\alpha} \\ +105\cdot R_z^5\cdot\mu_\alpha\cdot\Phi_{zzz\alpha} \\ +105\cdot R_z^5\cdot\mu_\alpha\cdot\Phi_{zzz\alpha} \\ +105\cdot R_z^5\cdot\mu_\alpha\cdot\Phi_{zzz\alpha} \\ -15\cdot R_z^5\cdot\mu_\alpha\cdot\Phi_{z\alpha\delta\delta} \\ -15\cdot R_z^5\cdot\mu_\alpha\cdot\Phi_{z\alpha\gamma\gamma} \\ -15\cdot R_z^5\cdot\mu_\alpha\cdot\Phi_{z\alpha\gamma\gamma} \\ -15\cdot R_z^5\cdot\mu_\alpha\cdot\Phi_{z\alpha\delta\delta} \\ -15\cdot R_z^5\cdot\mu_\alpha\cdot\Phi_{z\alpha\beta\beta} \\ -15\cdot R_z^5\cdot\mu_\alpha\cdot\Phi_{z\alpha\beta\beta} \\ -15\cdot R_z^5\cdot\mu_\alpha\cdot\Phi_{z\alpha\gamma\gamma} \\ -15\cdot R_z^5\cdot\mu_\alpha\cdot\Phi_{z\alpha\beta\beta} \\ -15\cdot R_z^5\cdot\mu_\alpha\cdot\Phi_{z\alpha\beta\beta} \\ -15\cdot R_z^5\cdot\mu_\alpha\cdot\Phi_{z\alpha\gamma\gamma} \\ -15\cdot R_z^5\cdot\mu_\alpha\cdot\Phi_{z\alpha\beta\beta} \\ -15\cdot R_z^5\cdot\mu_\alpha\cdot\Phi_{z\alpha\beta\beta} \end{bmatrix} \tag{3b}$$

Simplify Multipoles

$$T_{\alpha\beta\gamma\delta\epsilon}\mu_\alpha\Phi_{\beta\gamma\delta\epsilon} = R_z^{-11} \cdot \frac{1}{105} \cdot \begin{bmatrix} +105 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yzzz} \\ +105 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yzzz} \\ +105 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yzzz} \\ +105 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yzzz} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\delta\delta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\delta\delta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \end{bmatrix} \tag{4a}$$

$$\xRightarrow{\text{cleaned up:}} R_z^{-11} \cdot \frac{1}{105} \cdot \begin{bmatrix} -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\delta\delta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\delta\delta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \end{bmatrix} \tag{4b}$$

Exploit Traceless Conditions

$$T_{\alpha\beta\gamma\delta\epsilon}\mu_\alpha\Phi_{\beta\gamma\delta\epsilon} = R_z^{-11} \cdot \frac{1}{105} \cdot \begin{bmatrix} -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\delta\delta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\delta\delta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \end{bmatrix} \tag{5a}$$

$$\xRightarrow{\text{cleaned up:}} R_z^{-11} \cdot \frac{1}{105} \cdot \begin{bmatrix} -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\delta\delta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\delta\delta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \end{bmatrix} \tag{5b}$$

Reduce Indices

$$T_{\alpha\beta\gamma\delta\epsilon}\mu_\alpha\Phi_{\beta\gamma\delta\epsilon} = R_z^{-11}\cdot\frac{1}{105}\cdot\begin{bmatrix} -15\cdot R_z^5\cdot\mu_y\cdot\Phi_{yz\delta\delta} \\ -15\cdot R_z^5\cdot\mu_y\cdot\Phi_{yz\gamma\gamma} \\ -15\cdot R_z^5\cdot\mu_y\cdot\Phi_{yz\gamma\gamma} \\ -15\cdot R_z^5\cdot\mu_y\cdot\Phi_{yz\delta\delta} \\ -15\cdot R_z^5\cdot\mu_y\cdot\Phi_{yz\beta\beta} \\ -15\cdot R_z^5\cdot\mu_y\cdot\Phi_{yz\beta\beta} \\ -15\cdot R_z^5\cdot\mu_y\cdot\Phi_{yz\gamma\gamma} \\ -15\cdot R_z^5\cdot\mu_y\cdot\Phi_{yz\beta\beta} \\ -15\cdot R_z^5\cdot\mu_y\cdot\Phi_{yz\beta\beta} \\ -15\cdot R_z^5\cdot\mu_y\cdot\Phi_{yz\gamma\gamma} \\ -15\cdot R_z^5\cdot\mu_y\cdot\Phi_{yz\beta\beta} \\ -15\cdot R_z^5\cdot\mu_y\cdot\Phi_{yz\beta\beta} \end{bmatrix} \tag{6a}$$

$$\xRightarrow{\text{cleaned up:}} R_z^{-11}\cdot\frac{1}{105}\cdot\begin{bmatrix} -15\cdot R_z^5\cdot\mu_y\cdot\Phi_{yz\delta\delta} \\ -15\cdot R_z^5\cdot\mu_y\cdot\Phi_{yz\gamma\gamma} \\ -15\cdot R_z^5\cdot\mu_y\cdot\Phi_{yz\gamma\gamma} \\ -15\cdot R_z^5\cdot\mu_y\cdot\Phi_{yz\delta\delta} \\ -15\cdot R_z^5\cdot\mu_y\cdot\Phi_{yz\beta\beta} \\ -15\cdot R_z^5\cdot\mu_y\cdot\Phi_{yz\beta\beta} \\ -15\cdot R_z^5\cdot\mu_y\cdot\Phi_{yz\gamma\gamma} \\ -15\cdot R_z^5\cdot\mu_y\cdot\Phi_{yz\beta\beta} \\ -15\cdot R_z^5\cdot\mu_y\cdot\Phi_{yz\beta\beta} \\ -15\cdot R_z^5\cdot\mu_y\cdot\Phi_{yz\gamma\gamma} \\ -15\cdot R_z^5\cdot\mu_y\cdot\Phi_{yz\beta\beta} \\ -15\cdot R_z^5\cdot\mu_y\cdot\Phi_{yz\beta\beta} \end{bmatrix} \tag{6b}$$

Add up

$$T_{\alpha\beta\gamma\delta\epsilon}\mu_\alpha\Phi_{\beta\gamma\delta\epsilon} = R_z^{-11}\cdot\frac{1}{105}\cdot\begin{bmatrix} -30\cdot R_z^5\cdot\mu_y\cdot\Phi_{yz\delta\delta} \\ -60\cdot R_z^5\cdot\mu_y\cdot\Phi_{yz\gamma\gamma} \\ -90\cdot R_z^5\cdot\mu_y\cdot\Phi_{yz\beta\beta} \end{bmatrix} \tag{7}$$

Combining everything

$$T_{\alpha\beta\gamma\delta\epsilon}\mu_\alpha\Phi_{\beta\gamma\delta\epsilon} = \begin{bmatrix} -2/7\cdot R_z^{-6}\cdot\mu_y\cdot\Phi_{yz\delta\delta} \\ -4/7\cdot R_z^{-6}\cdot\mu_y\cdot\Phi_{yz\gamma\gamma} \\ -6/7\cdot R_z^{-6}\cdot\mu_y\cdot\Phi_{yz\beta\beta} \end{bmatrix} \tag{8}$$